

Data Analytics Internship - Task 3

Deep-Dive Analysis & Interactive Dashboard

Customer Segmentation and Sales Intelligence

Records	Unique Orders	Customers	Revenue
1,000	992	947	₹139,399,439.65

Analysis period: 01 Jan 2025 to 01 Jan 2026

1. Executive Summary

Objective. Task 3 extends the earlier EDA work into a deeper business analysis and an interactive BI dashboard. The selected deep-dive area is customer segmentation, supported by order frequency, customer revenue and recency.

Business question. Which customer groups contribute the most value, which groups are showing lower recent engagement, and where should business attention be focused?

Core KPIs

KPI	Formula	Business rationale
Total Revenue	SUM(Total_Sales)	Overall sales value generated.
Total Orders	DISTINCTCOUNT(Order_ID)	Measures order volume.
Total Customers	DISTINCTCOUNT(Customer_ID)	Measures unique customer reach.
Average Order Value	Total Revenue / Total Orders	Shows average revenue per order.
Total Units Sold	SUM(Quantity)	Measures sales volume in units.

KPI snapshot

Total Revenue: ₹139,399,439.65

Total Orders: 992

Unique Customers: 947

Average Order Value: ₹140,523.63

Total Units Sold: 5,435

2. Data & Analytical Method

The analysis uses the cleaned Task 1 sales dataset. The file contains 1,000 transaction rows and 15 fields covering orders, customers, products, categories, dates, quantities, prices and sales values.

For customer analysis, records are aggregated by unique Customer_ID only. Customer name and city are retained as descriptive attributes; revenue, order count and latest order date are calculated at customer level. This prevents duplicate customer rows from inflating customer counts.

Customer segmentation rules

Segment	Business rule
High Value	Total Revenue $\geq$ ₹250,000 and Total Orders $\geq$ 2
Loyal	Total Orders $\geq$ 2 and Recency $\leq$ 120 days, while not High Value
At Risk	Recency $>$ 180 days
New / Recent	Total Orders = 1 and Recency $\leq$ 90 days
Regular	All remaining customers

3. Sales Performance Deep Dive

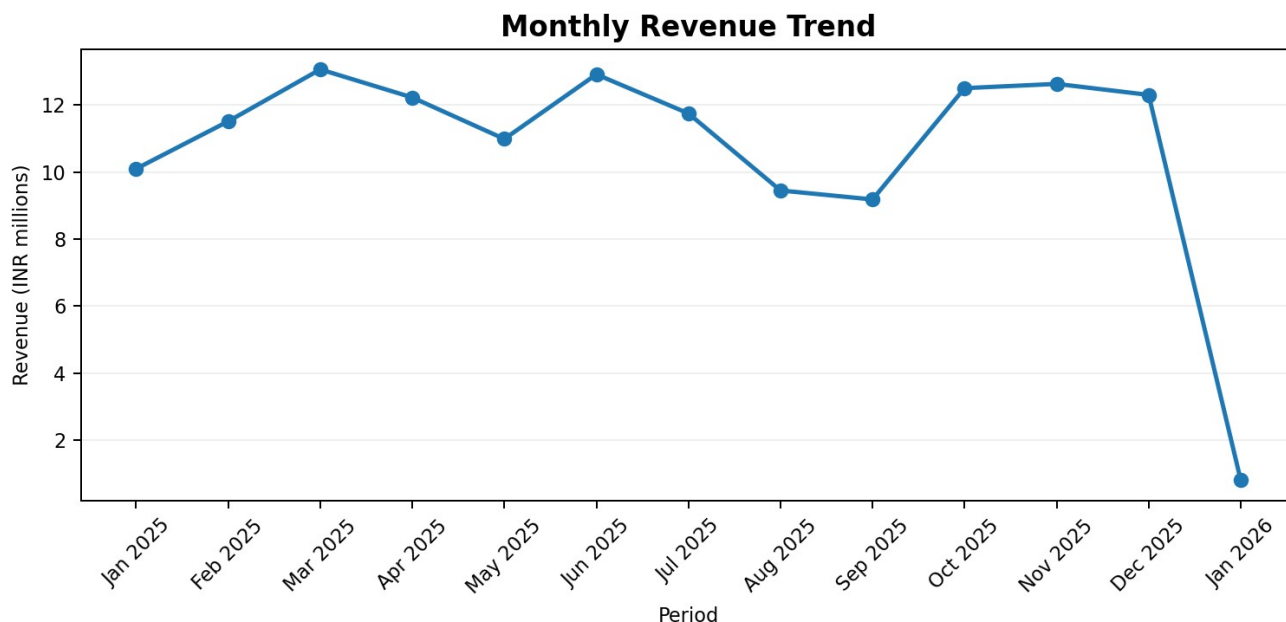


Figure 1. Monthly revenue trend. January 2026 is a partial month in the dataset and should not be compared directly with full months.

Peak full-month revenue occurs in March 2025 at ₹13,059,899.94. The lowest full month is September 2025 at ₹9,179,896.29. January 2026 contains only 809,389 revenue because the dataset ends on 1 Jan 2026.

Category concentration

Electronics is the largest category at ₹50,778,581.70, representing 36.4% of total revenue. The next largest categories are Education and Grocery.

4. Customer Segmentation Deep Dive

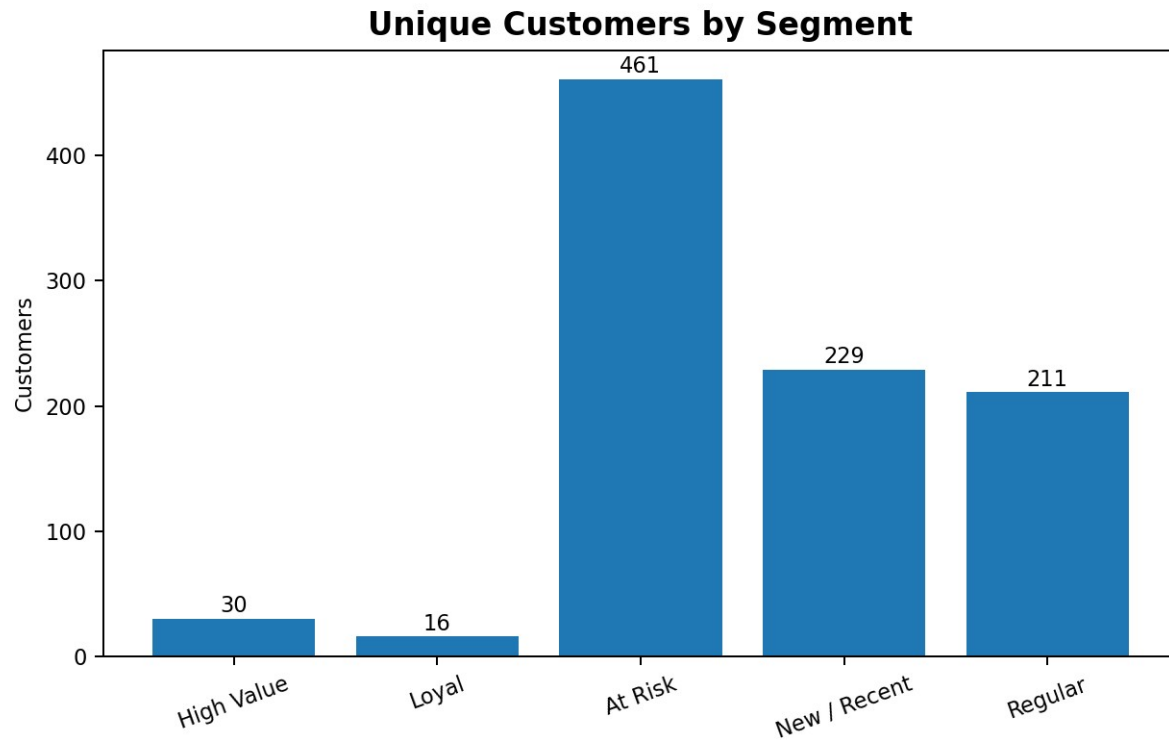


Figure 2. Distribution of unique customers across business-rule segments.

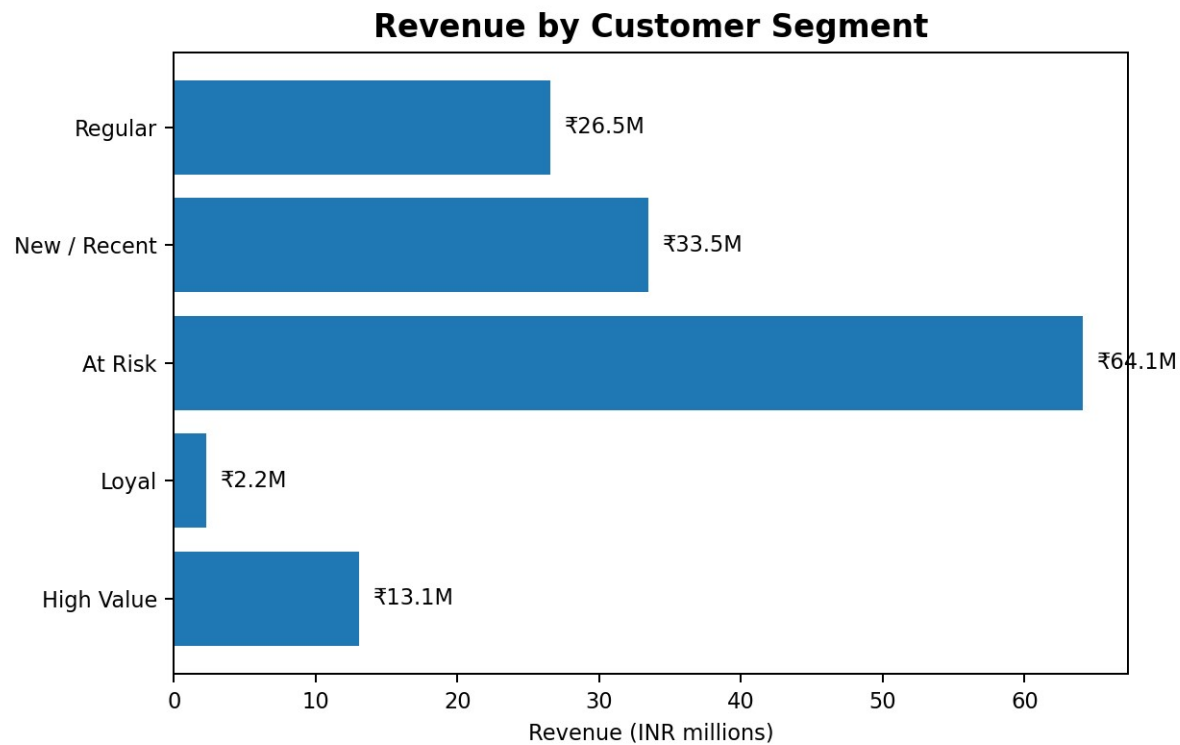


Figure 3. Revenue contribution by customer segment.

Interpretation

The At Risk segment contains 461 customers (48.7%) and contributes ₹64,100,751.66 (46.0%) of revenue. Its average recency is 271.9 days, so this is the clearest retention opportunity under the chosen rule.

New / Recent customers account for 229 customers (24.2%) and ₹33,476,902.43 (24.0%) of revenue. These customers have an average recency of 43.2 days and can be targeted for second-purchase conversion.

High Value customers are 30 customers producing ₹13,052,606.25; Loyal customers are 16 customers producing ₹2,248,355.49. Regular customers contribute ₹26,520,823.82. Segment definitions are business rules, not causal classifications.

5. Customer Value vs Frequency

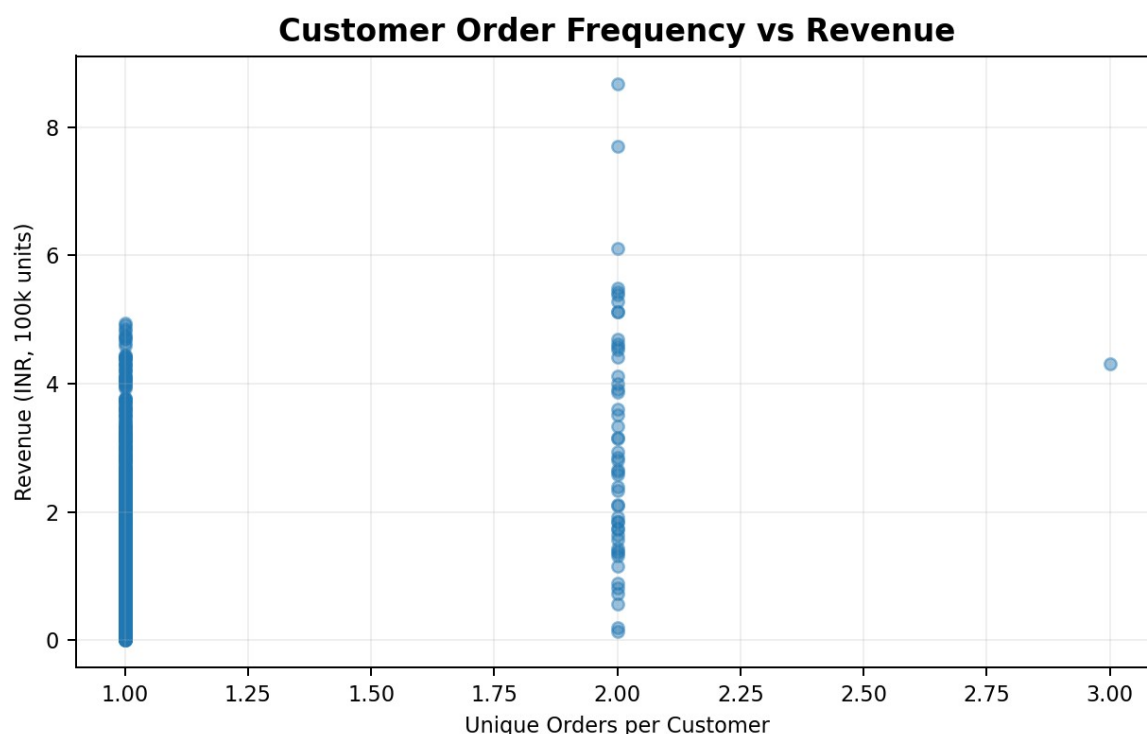


Figure 4. Customer order frequency versus revenue.

Most customers have one unique order, while a smaller group has two or more. This means repeat-order behavior is an important lever for increasing customer lifetime value. The analysis should be used as descriptive evidence rather than a causal conclusion.

6. Key Findings & Business Actions

Retention: At Risk customers represent nearly half of the unique customer base. Create reactivation campaigns, reminders and targeted offers for customers with long recency.

Second purchase: New / Recent customers form a sizable group. Build a post-purchase journey aimed at converting first-time buyers into repeat customers.

Value concentration: High Value customers are a smaller group but contribute meaningful revenue. Protect this cohort with tailored service, bundles and priority engagement.

Product/category focus: Electronics generates the largest category revenue. Use category-level promotions and cross-sell analysis while monitoring margins.

Seasonality: March 2025 is the strongest full month; September 2025 is materially lower. Compare campaigns, demand and inventory across those periods.

7. Interactive Dashboard Specification

The Power BI solution is structured into two pages:

Page	Contents
Sales Overview	Core KPI cards; revenue by category; monthly revenue trend; product revenue ranking; revenue by gender; category/city slicers.
Customer Deep Dive	Customer segment distribution; segment revenue; customer economics; segment-by-city analysis; customer detail table; segment and city slicers.

Interactivity

Recommended slicers are Category, City and Customer Segment. All KPI cards and charts should respond to slicer selections so the dashboard supports exploratory analysis.

8. Limitations & Data Notes

Customer segmentation is rule-based and descriptive. Thresholds can be tuned for a different business context. The dataset contains 1,000 rows, 992 unique orders and 947 unique customers, so some Order_ID and Customer_ID values repeat across rows. The reporting period includes a partial January 2026 observation; this should be treated carefully in month-over-month comparisons.